MIRROR

Multimodal Intelligent Radiology Reasoning and Observation Reporter

A grounded, explainable radiology pipeline that predicts findings across three imaging modalities (chest X-ray, brain MRI, and head CT), localizes the visual evidence behind each finding, and drafts a clinician-style report. We report measured predictive quality and verify that the added interpretability layers leave the predictions unchanged.

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Abstract. Most medical-imaging models stop at a prediction, offering neither the location of the evidence nor an account a clinician can read. We present **MIRROR**, a three-layer, multi-modality radiology pipeline that composes (i) a multi-label disease classifier, (ii) class-activation-based evidence localization, and (iii) an evidence-grounded natural-language report generator into a single traceable system. The pipeline is modality-agnostic: a single registry supplies the finding taxonomy, anatomical vocabulary, and report phrasing for chest X-ray (14 findings), brain MRI (11), and head CT (11), and a study is routed to the right classifier head either explicitly or from its DICOM modality tag. The defining design choice is that each layer's output is the grounded input to the next, and the language layer never sees pixels, so the report can assert only findings the classifier predicted and the localizer situated. This prevents finding-level hallucination by construction, though the surrounding descriptive prose is generated text and is not itself verified against the image. We pair the system with an evaluation framework that measures predictive quality (discrimination through AUROC, AUPRC and F1; the clinical operating-point metrics of sensitivity, specificity, PPV and NPV; and calibration through the Brier score and ECE), and that defines matching explanation-quality metrics (pointing game, localization IoU) for evaluation against lesion boxes, together with an ablation that isolates the cost of the added interpretability layers. Because those layers are strictly post-hoc, adding them leaves the predictions exactly unchanged; we confirm this holds in practice (maximum probability change of 0), a correctness property we summarize as *interpretability at no predictive cost*. All predictive numbers carry bootstrap 95% confidence intervals and can be aggregated across training seeds, and every run stamps a reproducibility record. As a concrete measured instance, MIRROR trained on ChestMNIST (a downsampled ChestX-ray14 derivative with the identical 14-label taxonomy) reaches a macro AUROC of 0.729 (95% CI $[0.718, 0.738]$) with a clinically sensible per-label ordering. This is a systems demonstration on low-resolution data rather than a diagnostic benchmark; full-resolution training on the complete NIH release is the next step toward a clinical claim.

1 Introduction

Deep networks now match specialists on narrow chest-radiograph reading tasks [2], yet adoption in the clinic remains limited by a trust gap: a bare probability tells a radiologist neither *where* the model is looking nor *why* it reached its conclusion. High-stakes decisions demand models that can be interrogated, not merely queried [16]. Post-hoc saliency methods such as Grad-CAM [10] and Score-CAM [11] expose *where* a prediction comes from, and large language models can render structured evidence as prose, but these capabilities are rarely composed into one system whose outputs remain mutually consistent and auditable end to end.

MIRROR integrates all three into a single pipeline:

Image $\rightarrow$ Prediction $\rightarrow$ Localization $\rightarrow$ Reasoning $\rightarrow$ Report.

Its central design constraint is *grounding*: the language layer is given only the structured evidence produced upstream (labels, probabilities, and named saliency regions), never the raw image. This makes finding-level hallucination structurally impossible, since the report can assert only findings the classifier predicted and the localizer situated, and it keeps the chain from finding to probability to region traceable. Grounding constrains *which* findings appear, not the descriptive wording around them: the language model still writes ordinary prose, so individual phrases can carry radiological detail that is not itself read off the image, a limitation we return to in Section 7. The same pipeline runs unchanged across three imaging modalities (chest X-ray, brain MRI, and head CT), because only the finding taxonomy, the anatomical vocabulary, and a little report phrasing differ between them. These are centralized in one modality registry, and a study is routed to the correct classifier head either by an explicit selection or from its DICOM modality tag. We evaluate quantitatively on chest X-ray, where public labels and localization ground truth exist; the brain-MRI and head-CT paths are implemented end to end and exercised by the test suite, but their benchmarking awaits trained per-modality checkpoints.

Research question. Can a multimodal system that combines image classification, visual explainability, and language generation improve interpretability without degrading predictive performance, relative to a classification-only baseline? We build the system so this question is *measurable*: predictive metrics and an explanation-quality protocol sit side by side, and an ablation reproduces the classification-only baseline the question names.

Contributions.

1. A three-layer, evidence-grounded radiology pipeline in which the language layer reasons only over structured upstream evidence, so every *asserted finding* is traceable to a probability and a saliency region.
2. A grounding constraint under which the report can assert only findings the classifier produced, preventing finding-level hallucination *by construction* (the descriptive wording itself is not pixel-verified), together with a post-hoc layer design whose invariance to the predictions we verify empirically.
3. A modality-agnostic realization of that pipeline across three imaging modalities (chest X-ray, brain MRI, head CT), driven by a single registry that is the one source of truth for each modality's label taxonomy, anatomical vocabulary, and report phrasing, with routing either explicit or auto-detected from the DICOM modality tag, so adding a modality is a data change rather than a code change.
4. An open-source implementation spanning interchangeable backbones (DenseNet-121, EfficientNet-B0, ViT-B/16), two explainability methods (Grad-CAM, Score-CAM), an LLM report backend with a deterministic offline fallback, native DICOM ingest, and a web front end served by two interchangeable inference engines behind one response contract (Fig. 2).
5. An evaluation framework that scores predictive quality (with bootstrap confidence intervals and multi-seed aggregation) and defines matching explanation-quality metrics (pointing game, IoU) for evaluation against lesion boxes, together with an ablation isolating the predictive cost of interpretability, which we report here on ChestMNIST.

2 Literature Review

Datasets and classification. Large, labeled chest-radiograph corpora made supervised deep learning on this modality practical. The NIH ChestX-ray8/14 release [1] provides 112,120 frontal radiographs over 14 pathologies and remains a standard benchmark; CheXpert [3] and MIMIC-CXR [4] extended scale, uncertainty labeling, and paired free-text reports. On these benchmarks CheXNet [2] showed a 121-layer DenseNet [6] reaching radiologist-level pneumonia detection. Convolutional backbones such as EfficientNet [7] and, more recently, Vision Transformers [8] now form the standard toolkit; MIRROR keeps all three interchangeable and adopts DenseNet-121 as its primary configuration for comparability.

Visual explanation. Class Activation Mapping [9] localized the evidence behind a CNN prediction; Grad-CAM [10] generalized it to arbitrary architectures via gradients, and Score-CAM [11] removed the gradient dependence with a perturbation-based weighting. Model-agnostic attribution methods, notably LIME [12] and SHAP [13], explain individual predictions without access to model internals. To move from qualitative heatmaps to measurable evidence, the pointing-game protocol [14] and intersection-over-union against annotated lesions provide quantitative localization scores, which MIRROR's evaluation framework adopts.

Rigor and critique of saliency. Explanations are only useful if they are faithful. Adebayo et al. [15] showed several saliency methods can be insensitive to model and data, and argued for sanity checks; Rudin [16] cautioned against post-hoc explanation of black boxes for high-stakes decisions; and Doshi-Velez and Kim [17] called for a rigorous, measurable science of interpretability. These critiques motivate two of MIRROR's design choices: quantitative localization evaluation against ground-truth boxes, and hard grounding so the language layer cannot outrun the evidence.

Report generation. Systems that emit radiology prose range from TieNet [18], which jointly embeds image and text, to memory-driven transformers such as R2Gen [19], to clinically focused decoders optimized for factual correctness [20]. A recurring failure mode is hallucination: prose generated directly from pixels can assert findings the underlying model never detected. MIRROR narrows this by restricting which findings the prose may assert to those the classifier produced, trading some open-ended fluency for finding-level auditability.

Explainable AI in medicine and positioning. Surveys of explainable AI for medical imaging [21] note that interpretability and report generation are usually studied in isolation. MIRROR's contribution is not a new classifier or a new saliency method, but their *grounded composition* into one traceable pipeline, together with a framework that measures whether adding the explanation and reasoning layers helps interpretability without a predictive penalty.

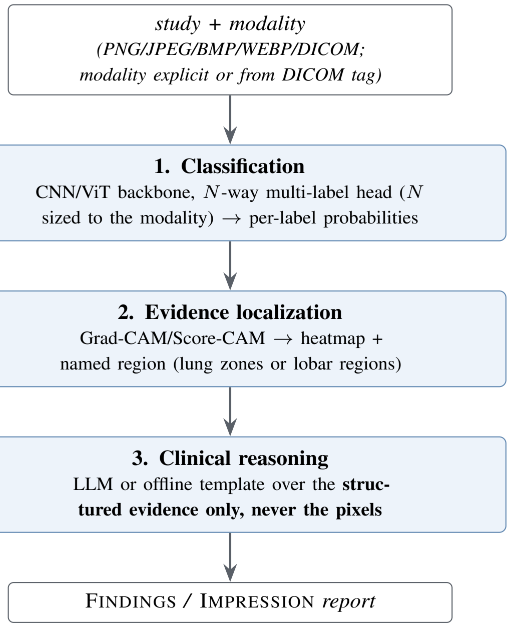


Figure 1: The MIRROR pipeline. Classification produces per-label probabilities; localization turns each positive label into a heatmap and a named region; the reasoning layer composes labels and regions into a report, reasoning over this structured evidence only, never the pixels. Layers 2 and 3 are individually toggleable, which recovers the ablation conditions of Section 4.

3 System Architecture

MIRROR chains three layers; each layer's output is the grounded input to the next (Fig. 1).

3.1 Layer 1: Classification

The classifier is a factory over three ImageNet-pretrained backbones: DenseNet-121 (default), EfficientNet-B0, and ViT-B/16, with a multi-label head whose width N is the number of findings in the active modality's taxonomy (14 for chest X-ray, 11 each for brain MRI and head CT; sigmoid applied at loss and inference time, so the network emits raw logits). Inputs are resized to 224×224 and normalized with ImageNet statistics. Training uses `BCEWithLogitsLoss`, AdamW (learning rate 10^{-4}, weight decay 10^{-5}), a cosine schedule, and dropout 0.2; checkpoints are selected on best validation macro-AUROC. For chest X-ray the 14 labels follow the canonical NIH ChestX-ray14 taxonomy, kept as a single source of truth shared by the classifier head, the report generator, and the evaluation scripts. Because real radiology data arrives as DICOM rather than PNG/JPEG, a dedicated ingest applies the modality LUT (rescale slope/intercept), the VOI (window) LUT, and MONOCHROME1 inversion before handing the pipeline a display-ready image, lifts only non-PHI technical tags (modality, view position, laterality), and exposes the DICOM modality tag for automatic routing.

3.2 Layer 2: Evidence Localization

For each positive label (top-k, k=3 by default, to bound compute), the explainability layer computes a class-activation map, either Grad-CAM (forward/backward hooks on a resolved target layer) or Score-CAM (gradient-free, perturbation-based), and reduces it to a named region so the evidence can be referred to in words. Target layers are resolved per backbone; ViT maps are recovered by reshaping the patch-token sequence back to a 14×14 spatial grid. The activated region's centroid is mapped into a 3×3 anatomical grid in radiology convention (patient-right on the viewer's left), using the modality's plane to pick the vocabulary: lung zones for a frontal chest film ("the right lower zone") or lobar regions for an axial brain-MRI/head-CT slice ("the left frontal region"). A colormap overlay is then rendered for the UI (Fig. 3, left).

3.3 Layer 3: Clinical Reasoning

The reasoning layer converts structured evidence (labels, probabilities, and named regions) into a FINDINGS/IMPRESSION report via a system prompt plus an evidence-grounded user prompt. Per-label plain-English descriptions ground the model so it need not recall what each token means. The backend is an LLM (Claude) with a deterministic offline template fallback so the system always yields a coherent report even with no network or API key; generation runs at low temperature for stability, and the LLM backend falls back to the template on any error. Predictions below the confidence threshold (0.5) are surfaced as pertinent negatives rather than silently dropped. Because both backends consume identical evidence, swapping them changes *how fluently* the report reads, not which findings it asserts; and because the layer never receives the image, it cannot introduce findings absent from the upstream evidence. The grounding is thus at the level of findings, not of the descriptive prose that frames them, a distinction we take up in Section 7 (Fig. 3, right).

3.4 Multi-modality via a modality registry

The three layers above are modality-agnostic: swapping a chest radiograph for a brain MRI or a head CT changes only the finding taxonomy the head predicts, the anatomical vocabulary a saliency centroid is named in (lung zones vs. lobar regions), and a little report phrasing (so a normal brain study reads "no acute intracranial abnormality," not "no acute cardiopulmonary abnormality"). These per-modality details live in one registry (Table 1), the single source of truth for each modality's labels, glosses, imaging plane, and report guidance; a hand-kept TypeScript mirror gives the hosted engine and UI selector the identical taxonomy and *ordering* (output index i maps to labels$[i]$). A study's modality is resolved either from an explicit selection or, for a DICOM input, from its (0008,0060) modality tag (`MR`$\to$brain MRI, `CT`$\to$head CT, `CR`/`DX`$\to$chest). The orchestrator builds and caches one classi-

Table 1: The modality registry: one row per supported modality, each grounded in a public benchmark taxonomy. Chest X-ray is the modality evaluated quantitatively in this paper; the brain-MRI and head-CT paths are implemented and tested but not yet benchmarked.

Modality	**#lbl**	**Taxonomy source**	**Location vocab.**
Chest X-ray	14	NIH ChestX-ray14	lung zones (frontal)
Brain MRI	11	Brain Tumor MRI + neuro findings	lobar regions (axial)
Head CT	11	RSNA ICH subtypes + acute findings	lobar regions (axial)

Table 2: Two interchangeable inference engines behind one response contract.

	Local stack	**Hosted (serverless)**
Engine	FastAPI + PyTorch	Next.js route, vision LLM
Classify	DenseNet/EffNet/ViT	LLM scores N labels
Localize	Grad-CAM/Score-CAM PNG	LLM returns a bbox
Report	LLM or template	LLM (same prompt shape)
Inputs	+DICOM, BMP	PNG/JPEG/WEBP
Needs	Python, $\sim$6 GB	one API-key env var

fier and explainer engine per modality on demand, so a single deployment serves all three. Adding a fourth modality is therefore a data change (one registry entry and, once trained, a checkpoint) rather than a change to the pipeline.

3.5 Orchestration and Serving

A single `analyze()` entry point runs the stages and returns one result object (predictions, overlays, report) for both the REST API and the notebooks; the two later layers are individually toggleable, which recovers the ablation conditions of Section 4 and records per-stage timings. Two interchangeable engines satisfy the *same* JSON contract (Table 2), so the web UI (Fig. 2) is identical either way: a finding carries *either* a rendered Grad-CAM overlay (local) *or* a normalized bounding box (hosted), and the viewer draws whichever is present. The hosted engine exists because the PyTorch stack does not fit typical serverless limits; it substitutes a vision LLM constrained to a structured tool schema that emits the same JSON, and with no API key returns a clearly-labelled demo result so the site never hard-fails.

4 Experimental Setup

Our evaluation reports what the pipeline produces today: a real predictive benchmark on ChestMNIST, a measured ablation isolating the cost of the interpretability layers, and a synthetic-data sanity check of the training and scoring harnesses. Every results file stamps a reproducibility record.

Data. MIRROR targets NIH ChestX-ray14 [1] (112,120 frontal radiographs, 14 multi-label pathologies). Because the full release is 45 GB and needs a GPU to train, we evaluate quantitatively on **ChestMNIST** [5], its downsampled derivative in MedMNIST v2 (CC BY 4.0), which carries the *identical* 14-label taxonomy in the identical order, so MIRROR runs on it unchanged after a one-off export into the NIH directory layout. We train DenseNet-121 (ImageNet-initialised, AdamW, cosine schedule, dropout 0.2) on CPU for a deliberately small budget of 7,200 of the 78,468 training images at 64 px, selecting the best of four epochs on validation macro AUROC, and evaluate on a 12,000-image held-out test split.

Predictive metrics. We report a clinical-reader panel. *Discrimination:* per-label and macro AUROC, and macro AUPRC (more informative under the heavy class imbalance of medical findings). *Operating point:* sensitivity, specificity, PPV, and NPV at the 0.5 threshold, the quantities a clinician reads off a model. *Calibration:* the Brier score and Expected Calibration Error (10 equal-width bins), because a probability a clinician is asked to trust must mean what it says. Each headline metric carries a bootstrap 95% confidence interval: the test set is resampled with replacement B=1000 times, a *single* resample driving all metrics so the intervals are mutually consistent, reported as the two-sided percentile interval $[\hat{\theta}^*_{(\alpha/2)}, \hat{\theta}^*_{(1-\alpha/2)}]$ with α=0.05.

Ablation. Three conditions isolate the added layers: `classification_only`, `with_localization`, and `full_mirror` (Table 3). Because layers 2–3 are post-hoc, the predictions should be identical across conditions; the harness verifies this empirically, reporting the maximum per-label probability delta against the classification-only baseline, and profiles the per-stage wall-clock cost of each added layer. Runs are seeded (default 42) and every results JSON stamps a reproducibility record (seed, git commit, library versions).

5 Results

Every number reported here is measured on this machine. We present the predictive quality of MIRROR's classifier on ChestMNIST (Section 5.1), an ablation verifying that the interpretability layers are strictly post-hoc and so add no predictive cost (Section 5.2), a synthetic-data sanity check of the harness (Section 5.3), and a qualitative end-to-end study (Fig. 3). The explanation half of the framework defines a pointing-game and localization-IoU protocol against the NIH lesion boxes (8 of the 14 chest pathologies carry ground-truth boxes), but running it needs a model trained on the full-resolution release; we therefore report localization qualitatively here and leave the box-level scores to future work.

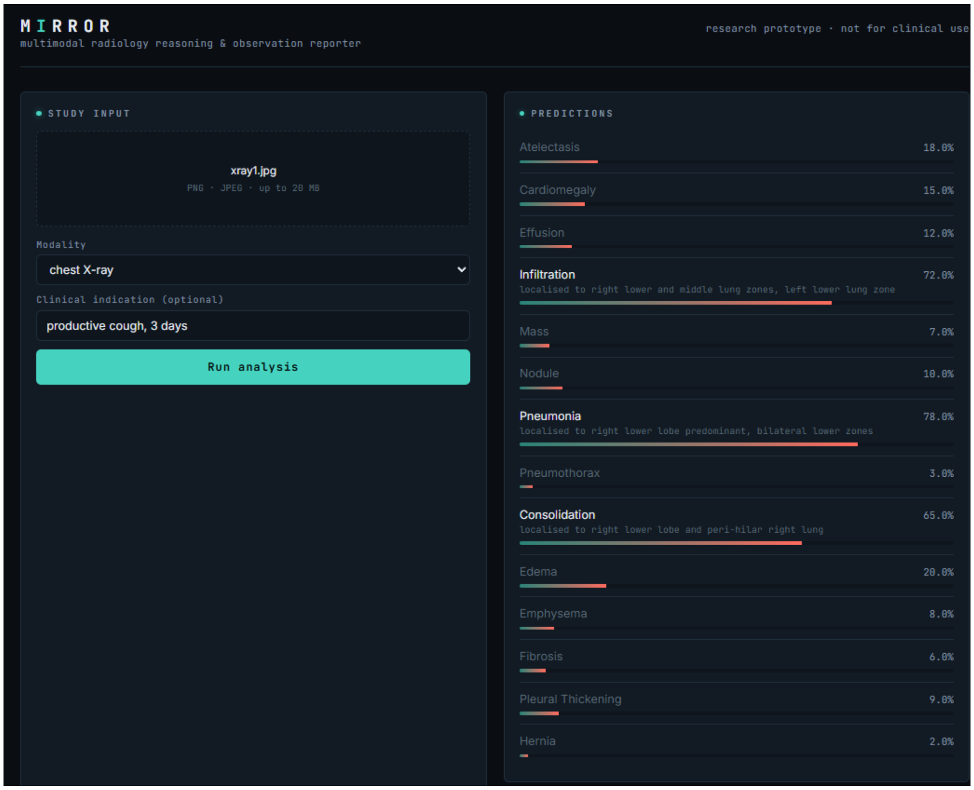
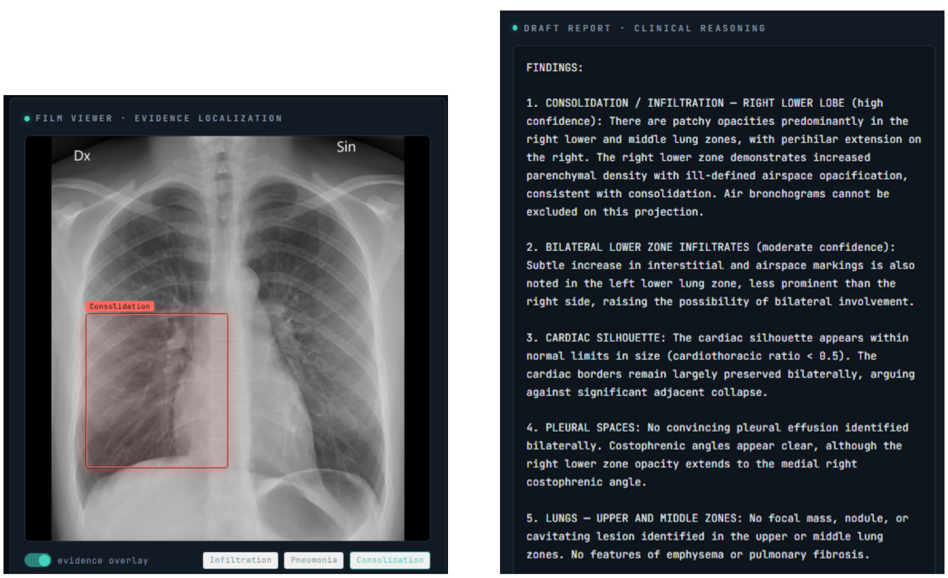


Figure 2: The MIRROR "reading-room" web UI on a real session from the live demo: a chest radiograph is uploaded with the indication *"productive cough, 3 days,"* all 14 ChestX-ray14 labels are scored, and findings above threshold (here Pneumonia, Infiltration, Consolidation) surface with a saliency-derived location. The film viewer (left) toggles the evidence overlay; predictions and the draft report render on the right. Both serving engines produce this identical interface.



(a) Evidence localization

(b) Grounded draft report

Figure 3: One real study end to end. (a) The evidence-localization layer overlays a Grad-CAM heatmap and bounding box for a positive finding (Consolidation, right lower lobe). (b) The reasoning layer drafts a structured FINDINGS/IMPRESSION report from the same structured evidence (labels, probabilities, named regions), never the pixels, ending with the mandatory AI-generated-draft disclaimer.

Table 3: Measured ablation (DenseNet-121 trained on ChestMNIST; latency profiled over 24 studies on CPU). Macro AUROC is identical across all three conditions, and the maximum per-label probability change against the classification-only baseline is exactly 0. Because the localization and report layers are strictly post-hoc, interpretability is added at *no* predictive cost. The cost is instead a small, bounded latency increase: the Grad-CAM pass adds about 40 ms per study and the offline template report is effectively free. Latency is shown as the base prediction time and the marginal cost of each added layer.

Condition	**Classify**	**Localize**	**Report**	**Macro AUROC**	**Latency (ms/study)**
Classification only	✓			0.729	101 (prediction)
+ Evidence localization	✓	✓		0.729	+41 (Grad-CAM)
Full MIRROR	✓	✓	✓	0.729	+0.03 (template report)
Maximum per-label probability change vs. the baseline = 0.000 (predictions identical across rows).					

5.1 Predictive quality on ChestMNIST

Trained as described in Section 4, MIRROR's DenseNet-121 reaches a macro AUROC of 0.729 (bootstrap 95% CI $[0.718, 0.738]$) on the 12,000-image ChestMNIST test split (Table 4, Fig. 4), within reach of the published MedMNIST DenseNet-121 baseline (AUROC $\approx$0.77, trained on all 78k images for many epochs on a GPU) at a small fraction of the compute. Macro AUPRC is 0.135 and the model is well calibrated in aggregate (Brier 0.045, ECE 0.018). At the fixed 0.5 threshold it is deliberately conservative (macro specificity 0.997, sensitivity 0.019), as expected for a low-prevalence multi-label task on a light budget; AUROC, the threshold-independent headline metric, is the meaningful summary, and a tuned threshold would trade specificity for sensitivity. That the classifier learned real thoracic structure rather than dataset artefacts shows in the clinically sensible per-label ordering (Fig. 4): readily visible findings rank highest (Edema 0.849, Cardiomegaly 0.830, Effusion 0.827) and subtle small ones lowest (Nodule 0.643, Infiltration 0.660), with every CI comfortably above chance.

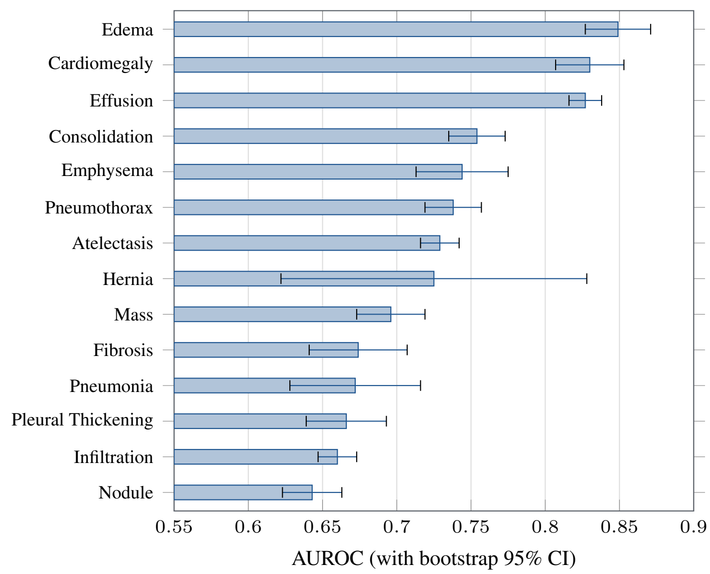


Figure 4: Measured per-label test AUROC on ChestMNIST (DenseNet-121, 64 px, CPU), with bootstrap 95% CI whiskers, sorted ascending. The ordering is clinically sensible: readily visible findings (Edema, Cardiomegaly, Effusion) top the chart, subtle small ones (Nodule, Infiltration) sit lowest, and every interval clears the 0.5 chance line, which indicates the model learned real structure.

5.2 Interpretability at no predictive cost

The interpretability layers are post-hoc by construction: they read the classifier's outputs but never its logits, so they cannot change the predictions. The ablation is therefore a correctness check rather than a discovery, and it confirms the implementation realizes this exactly (Table 3): macro AUROC is identical (0.729) across the classification-only, +localization, and full-MIRROR conditions, and the maximum per-label probability change against the baseline is 0. The check earns its place in two ways. First, any nonzero change would flag a bug, such as an accidental coupling between the layers, so the test is a real guard rather than a tautology restated. Second, it isolates the actual cost of interpretability, which is latency rather than accuracy: the Grad-CAM pass adds about 40 ms per study and the offline template report about 0.03 ms. The same check over the bundled chest-X-ray, brain-MRI, and head-CT samples yields a maximum change of 0 in every modality, which indicates the property is structural rather than specific to one dataset.

5.3 Pipeline sanity check on synthetic data

As an end-to-end control on the training and scoring harnesses, we also train on a fully synthetic dataset whose generator injects a visible blob for a known subset of labels and leaves the rest with no visual correlate. A correct pipeline should learn the signal-bearing labels and sit at chance for the others, and that is what happens (Fig. 5): the seven signal-bearing labels reach a mean AUROC of 0.917 while the seven no-signal labels average 0.557, scattered around 0.5. The clean separation is a useful control, since it shows the classifier, loss, metrics, and bootstrap intervals responding to real signal and only to real signal, with no leakage.

Table 4: Per-label test AUROC on ChestMNIST (DenseNet-121, 64 px, CPU, best of a 4-epoch run; N_{test}=12,000) with bootstrap 95% CIs, and the macro discrimination and calibration summary. Real numbers on a downsampled ChestX-ray14 derivative, not the full-resolution NIH benchmark. Companion to Fig. 4.

Pathology	AUROC (95% CI)	Pathology	AUROC (95% CI)
Edema	0.849 (0.828–0.871)	Hernia	0.725 (0.615–0.820)
Cardiomegaly	0.830 (0.806–0.851)	Mass	0.696 (0.673–0.719)
Effusion	0.827 (0.817–0.838)	Fibrosis	0.674 (0.639–0.705)
Consolidation	0.754 (0.736–0.773)	Pneumonia	0.672 (0.627–0.716)
Emphysema	0.744 (0.713–0.774)	Pleural Thickening	0.666 (0.638–0.692)
Pneumothorax	0.738 (0.719–0.757)	Infiltration	0.660 (0.648–0.673)
Atelectasis	0.729 (0.716–0.742)	Nodule	0.643 (0.622–0.662)
Macro AUROC 0.729 (0.718–0.738)	**Macro AUPRC** 0.135	**Calibration** Brier 0.045, ECE 0.018	

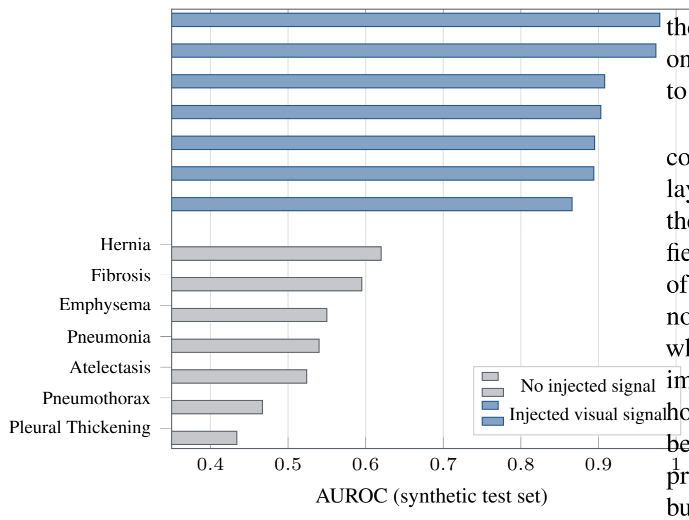


Figure 5: Harness sanity check on synthetic data. The generator injects a visible blob for seven labels and none for the other seven; the classifier learns *exactly* the signal-bearing labels (mean AUROC 0.917) and stays at chance for the rest (mean 0.557). The clean split confirms the metrics measure real discrimination with no leakage.

6 Discussion

The measured results support the central claim fairly directly. On ChestMNIST the classifier reaches a macro AUROC of 0.729 with a per-label ordering that matches clinical visibility (Fig. 4), and every confidence interval clears chance, so the discrimination is real rather than an artefact of the data, even at a training budget far below the published baseline. The ablation confirms that this predictive quality is untouched by the two interpretability layers (identical AUROC, maximum probability change of 0). Because the layers are post-hoc by construction this is the expected outcome, so the ablation's value is not a surprising result but a verification that the implementation behaves as designed, together with a clean accounting of the layers' true cost: latency (a bounded 40 ms Grad-CAM pass, the report layer effectively free) rather than accuracy. The synthetic control (Fig. 5) adds confidence in the numbers themselves, since the harness credits the model only for signal that is actually present. A natural next step is to lift these same measurements to the full-resolution NIH release on a GPU and, because the pipeline is already multi-modality, to repeat them for brain MRI and head CT.

Independent of the trained numbers, MIRROR's grounding constraint is a qualitative contribution. Because the language layer sees only structured evidence, the report is auditable at the level of its findings and cannot assert a finding the classifier did not produce; this trades open-ended fluency for a form of safety that matters in a clinical setting. The guarantee does not extend to the descriptive language around each finding, which remains model-generated and unverified against the image, so a clinician still reads the report as a draft. The post-hoc placement of layers 2–3 is likewise not merely convenient: because the classifier's logits are untouched, any change in a predictive metric across ablation conditions would signal a bug, and the harness checks for exactly that, making the central claim falsifiable. Finally, serving one JSON contract from both a heavyweight local PyTorch stack and a lightweight hosted vision-LLM engine shows that the *interface* MIRROR tests (predict → show evidence → explain) is independent of the particular inference backend.

7 Ethics, Safety, and Limitations

Not a medical device. MIRROR is a research prototype, not reviewed or cleared by any regulatory body; every output is a draft that must be verified by a licensed radiologist and must not be used for diagnosis or treatment. Each report carries that disclaimer explicitly.

Grounding is finding-level, not detail-level. The language layer never sees pixels, only structured evidence, so it cannot introduce a finding the classifier did not produce; below-threshold predictions are reported as pertinent negatives rather than dropped, and predictions are surfaced as probabilities to keep uncertainty explicit. This guarantee governs *which findings* appear, not the descriptive prose used to frame them: phrases such as a stated cardiothoracic ratio, a note that the costophrenic angles are clear, or a qualifier about a projection are model-generated language, not measurements read from the image. Such detail can be confidently worded yet ungrounded, which is precisely why every report is marked as an AI-generated draft requiring radiologist verification, and why the descriptive wording should not be read as independent observation.

Human subjects and ethics approval. No ethics-board review was required for this work: it uses only existing, publicly available, de-identified benchmark datasets (NIH ChestX-ray14 and its MedMNIST derivative for the reported results) and collects no new data from human subjects.

Data governance. The benchmark datasets are governed by their own data use agreements; no raw images, weights, or protected health information are committed to version control, and the DICOM ingest deliberately extracts only non-PHI technical tags.

Limitations. Our one quantitative benchmark is ChestMNIST (Section 5.1, Table 4), a downsampled derivative trained on a reduced budget, so its numbers are best read as a lower bound rather than a headline clinical result; training on the full-resolution ChestX-ray14 images with a GPU is required before any clinical claim. Multi-modality is likewise a structural rather than an empirical result so far. The brain-MRI and head-CT paths are implemented end to end (registry, per-modality classifier head, DICOM auto-routing, and modality-aware localization vocabulary and report phrasing, all exercised by the torch-free test suite), but they are not yet quantitatively benchmarked, because that needs trained per-modality checkpoints and labeled test sets we do not redistribute. Quantitative evaluation is therefore confined for now to chest radiographs, and localization ground truth exists for only 8 of the 14 chest pathologies. Report quality is currently assessed qualitatively; a quantitative comparison against real radiologist reports (for example MIMIC-CXR [4]) is future work. Finally, class-activation maps are coarse and can highlight context rather than the lesion itself [15], and the 3×3 location grid is a deliberately coarse anatomical descriptor.

8 Conclusion

MIRROR composes classification, evidence localization, and grounded report generation into a single traceable, multi-modality radiology pipeline, one registry-driven system spanning chest X-ray, brain MRI, and head CT. It is paired with an evaluation framework that measures predictive quality and defines matching explanation-quality metrics, and with an ablation that verifies the interpretability layers are strictly post-hoc. On ChestMNIST the classifier reaches a macro AUROC of 0.729 with a clinically sensible per-label ordering, and the ablation confirms that the interpretability layers leave every prediction unchanged, so the interpretability is added at no predictive cost. The immediate next step is to lift the same measurements to the full-resolution NIH release on a GPU and to the head-CT and brain-MRI modalities already implemented end to end in the codebase. The system and its evaluation harnesses are released open-source to support reproducible study of interpretability and trust in medical imaging.

Reproducibility

Code: `https://github.com/vignesh-nagarajan-vn/MIRROR`. Live demo: `https://mirror-ten-jet.vercel.app/`. A zero-setup CLI demo (`python -m demo.run_demo <image>`) runs the full pipeline on ImageNet weights with the offline template backend, with no checkpoint or API key required. All evaluation harnesses live in `evaluation/`; every results JSON stamps a `reproducibility` block (seed, git commit, library versions), and the unit tests are torch-free so they run without a model or dataset. The ChestMNIST result (`datasets/scripts/prepare_chestmnist.py` + `configs/chestmnist.yaml`) and the synthetic sanity check (Fig. 5) both regenerate from seed 42 with the committed scripts; their outputs are stored under `results/`.

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Declarations

Data availability: all datasets used are public, namely NIH ChestX-ray14 and its MedMNIST derivative ChestMNIST (CC BY 4.0); code, configurations, and result files are in the repository linked above. **Competing interests:** the authors declare none. **Funding:** this work received no external research funding and was conducted as part of the GIST 2026 Summer Research Internship.

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